Problem Set #4 - Answer Sheet

1. $\frac{5}{324}$

$$p = \binom{6}{1,1,1,1,1,1}(\frac{1}{6})^6$$

2.

a. $f(X = x) = 2x$

$$f(X = x) = \int_0^x f(X = x, Y = y)dy = \int_0^x \frac{4y}{x}dy = 2x$$

b. $f(Y = y) = -4ylny$

$$f(Y = y) = \int_y^1 \frac{4y}{x}dx = -4ylny$$

c. $E[X] = \frac{1}{3}(1 - y^3)$

$$E[X] = \int_y^1 2x^2dx = \frac{1}{3}(1 - y^3)$$

3. 0.903

Let

p_1: The probability that the number of visitors exceeds 2000 in one week

Based on the information given in the problem, we obtain:

$$p_1 = 1 - F(x = 2000) = 1 - \Phi(\frac{2000 - 2200}{\sqrt{52900}}) = 0.807$$

Therefore, we can compute the probability that the weekly number of visitors ecxeeds 2000 in at least 2 of the next 3 weeks is

$$p = \binom{3}{2}p_1^2(1 - p_1) + p_1^3 = 0.903$$

4. 0.1416

Let

T: A random variable represents that the baby is tired
RT: A random variable represents the time in minutes until the baby rubs her eyes

Based on the information given in the problem, we obtain:

$$P(T) = \frac{3}{4}$$

When the baby is tired: $RT \sim Exp(\lambda = 3)$

When the baby is not tired: $RT \sim Exp(\lambda = 1)$

We can further compute that

$$P(T \mid RT = 2) = \frac{P(T, RT = 2)}{P(RT = 2)} = \frac{P(RT = 2 \mid T)P(T)}{P(RT = 2 \mid T)P(T) + P(RT = 2 \mid \overline{T})P(\overline{T})}$$

Because RT is a continuous random variable, therefore $P(RT = 2 \mid T) = f(RT = 2 \mid T)\epsilon$

Then we can transform the above equation and compute the updated belief

$$P(T \mid RT = 2) = \frac{P(T, RT = 2)}{P(RT = 2)} = \frac{P(RT = 2 \mid T)P(T)}{P(RT = 2 \mid T)P(T) + P(RT = 2 \mid \overline{T})P(\overline{T})} = \frac{f(RT = 2 \mid T)P(T)\epsilon}{f(RT = 2 \mid T)P(T)\epsilon + f(RT = 2 \mid \overline{T})P(\overline{T})\epsilon} = \frac{f(RT = 2 \mid T)P(T)}{f(RT = 2 \mid T)P(T) + f(RT = 2 \mid \overline{T})P(\overline{T})} = 0.1416$$

5.
a. $Beta(9,4)$

b. 0.927

$$p = 1 - F(0.5) = 0.927$$

6 - 8.

See the code file

9.
a. 0.84134

$$1 - F(103) = \Phi(\frac{103 - 101}{\sqrt{4}}) = \Phi(1) = 0.84134$$

b. 0.0145

$$P(F=1\,|\,Temp=100)=\frac{P(Temp=100\,|\,F=1)P(F=1)}{P(Temp=100)}=\frac{f_1(100)P(F=1)}{f_1(100)P(F=1)+f_2(100)P(F=0)}=0.0145$$

where f_1, f_2 are the PDF of T when $F=1$ and $F=0$, seperatly.

c.

See the code file

d. $8.5594 * 10^{-39}$

Let

N: A random variable represents the number of samples that have a temperature within 0.1 degrees of 100 degrees

p: The peobability that the temperature is within 0.1 degrees of 100 degrees

Therefore

$$P(K<20)=\sum_{k=0}^{20}\binom{10000}{k}p^k(1-p)^{10000-k}$$

Because N is large, and p is small, then we can approximate the distribution of K as Poisson distribution

$$K\sim Poi(145)$$

And then we can approximate that

$$P(K<20)=F(20)=8.5594*10^{-39}$$

10.

a. $Beta(4,3)$

b. c.

See the code file